DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot C

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code.

Series SP-C04: Cafe orders [20 marks]

Scenario: You are the analyst for a campus cafe. The series orders (Orders per day) is recorded daily. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 7 observations. Run the code below to create the series x and answer the tasks.

Evidence: run this code in R (do not edit it).

```
set.seed(6212)
m <- 7                      # observations per cycle (daily data)
n <- m * 10                 # 10 full cycles
t <- 1:n
trend <- 161.0 + 0.808 * t
orders <- round(trend + 36.8 * cos(2 * pi * (t - 2) / m) + rnorm(n, 0, 9.4), 0)
x <- ts(orders, frequency = m, start = c(1, 1))
```

Tasks:

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(orders ~ 1:n) and report the slope per day). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 7 seasonal values). Name the peak day and the trough day with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 10 cycles is the highest value in the peak day from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(orders ~ 1:n))) and compute acf(d, lag.max = 7, plot = FALSE) using observation lags. Report the autocorrelation at lag 7 and at lag 3, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4]
5. Additive or multiplicative? Compute the range (max minus min) of orders in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per day [2]. Write one sentence of advice to management based on your findings [2]. [6]

Remember: *seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative.*

Code of Conduct: I certify that this work is my own. Signature: _____

Activity - 8

1. Look at the series

```
set.seed(6212)
m <- 7 ; n <- 70 ; t <- 1:n
orders <- round(161 + .80*t + 36.8*cos(2*pi*(t-2)/m) +
                rnorm(n, 0, 9.4))
x <- ts(orders, frequency = 7)
plot(x, main = "cafe daily orders")
coef(lm(orders ~ t))
```

Ans observation = 70 , cycle length = 7 days. full cycle= the trend is rising. Report the slope from the output.

2. Seasonal Pattern.

```
dc <- decompose(x)
s <- dc$figure
print(round(s, 2))
which.max(s); which.min(s)
max(s) - min(s)
```

Ans The peak day is the position with the max seasonal value. The through day is the position with the min seasonal value. seasonal swing = Peak - trough.

it consistent.

```
library(ggplot2)
df <- data.frame(orders = as.numeric(x), day = 1:70,
                 position = cycle(x), cycle = rep(1:10, each
ggplot(df, aes(position, orders, group = cycle, colour
               = factor(cycle))) +
geom_line() + labs(colour = "cycle", x = "Day in cycle")
```

Ans The code counts how many cycles have their highest order on the seasonal peak day. Similar repeating lines indicate a consistent Pattern.

4. Autocorrelation

```
d <- resid(lm(orders ~ t))
a <- acf(d, lag.max = 7, plot = FALSE)
a$acf[8]
a$acf[4]
2/sqrt(n)
```

Ans compare lag 7 & lag 3 autocorrelations with the significance bound $2/\sqrt{70} \approx 0.239$.

5. Additive or multiplicative

```
r1 <- diff(range(orders[1:7]))
r10 <- diff(range(orders[64:70]))
r10/r1
dc <- decompose(x, type = "additive")
adjusted <- orders - dc$seasonal
coef(lm(adjusted ~ t))
```

Ans compare the first & last cycle ranges. If the ration is near 1, additive seasonality is reasonable.